

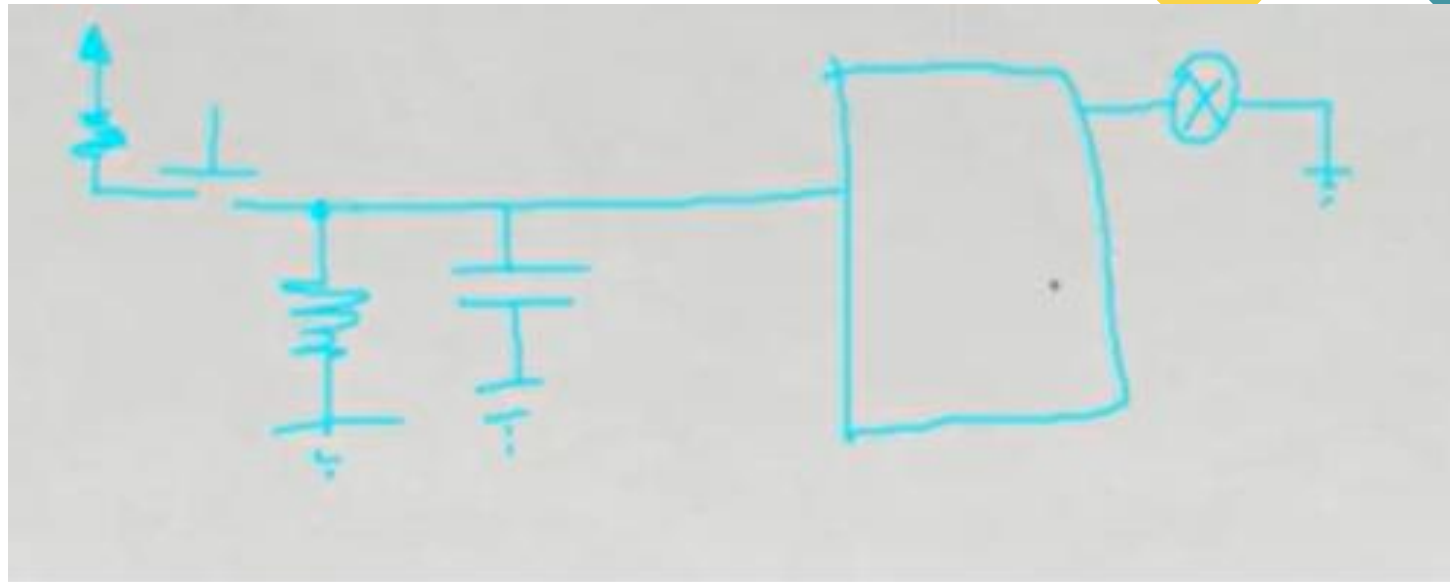


# **Decodificador BCD para display de 7 seg**

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# Introdução

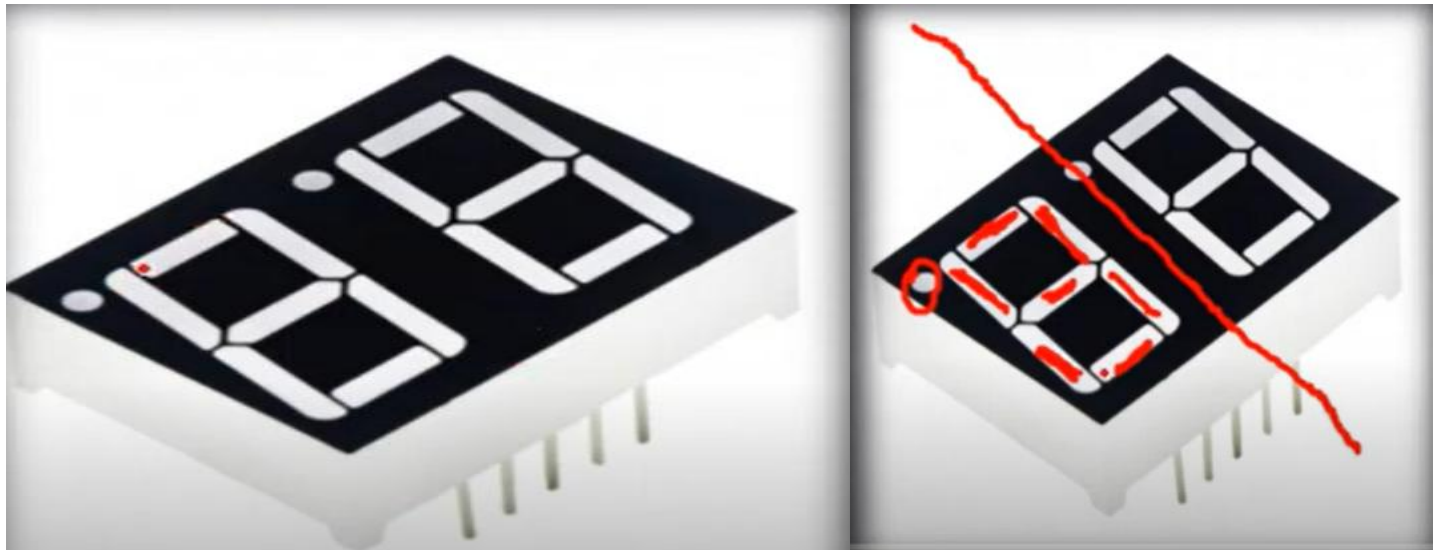
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# Dispositivos



- **Led**
  - São diodos
- **Diodos**
  - Há diversos tipos
- **Display**
  - Composto de sete leds



# Led para cada numero

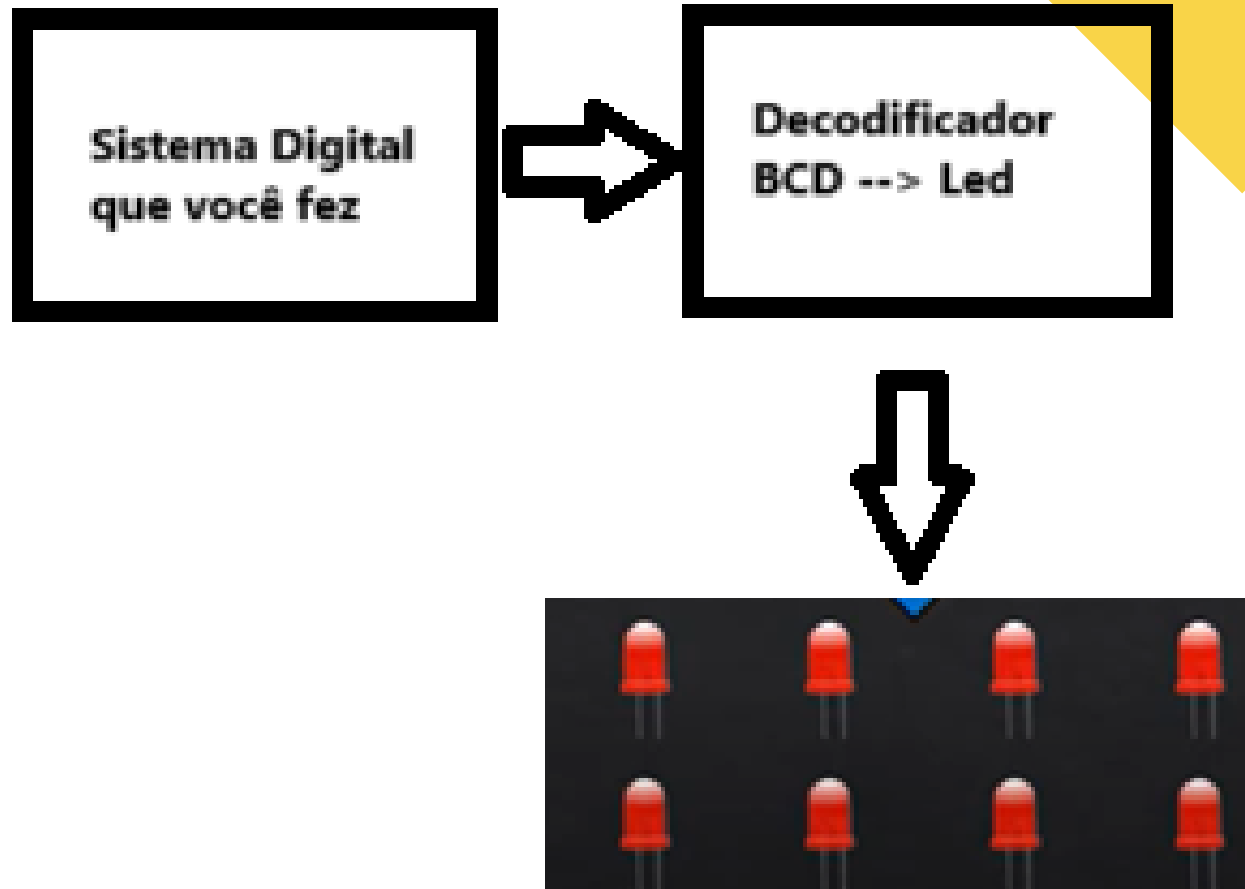
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Sistema digital  
que você fez



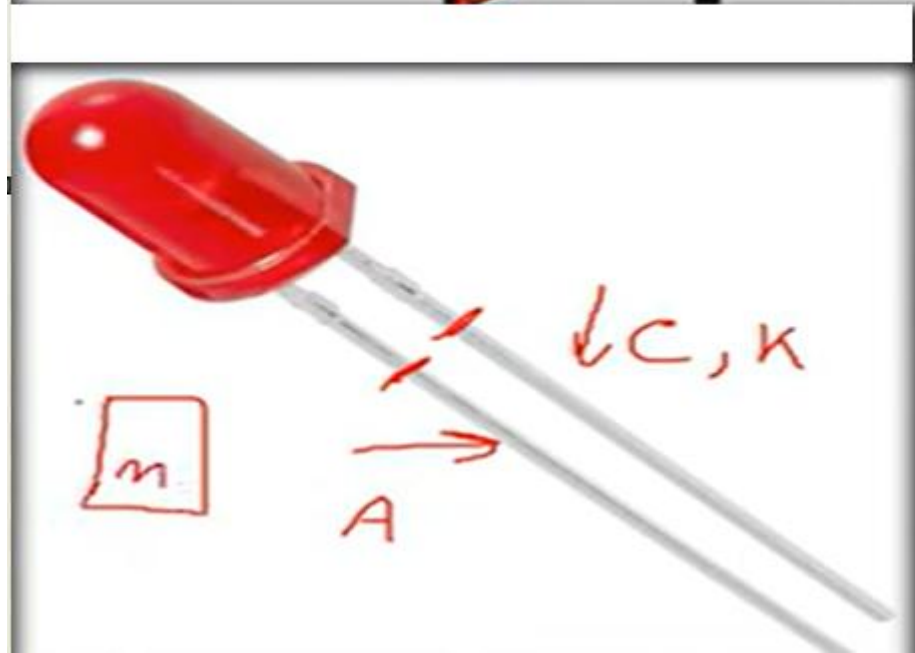
# Decodificador 8421

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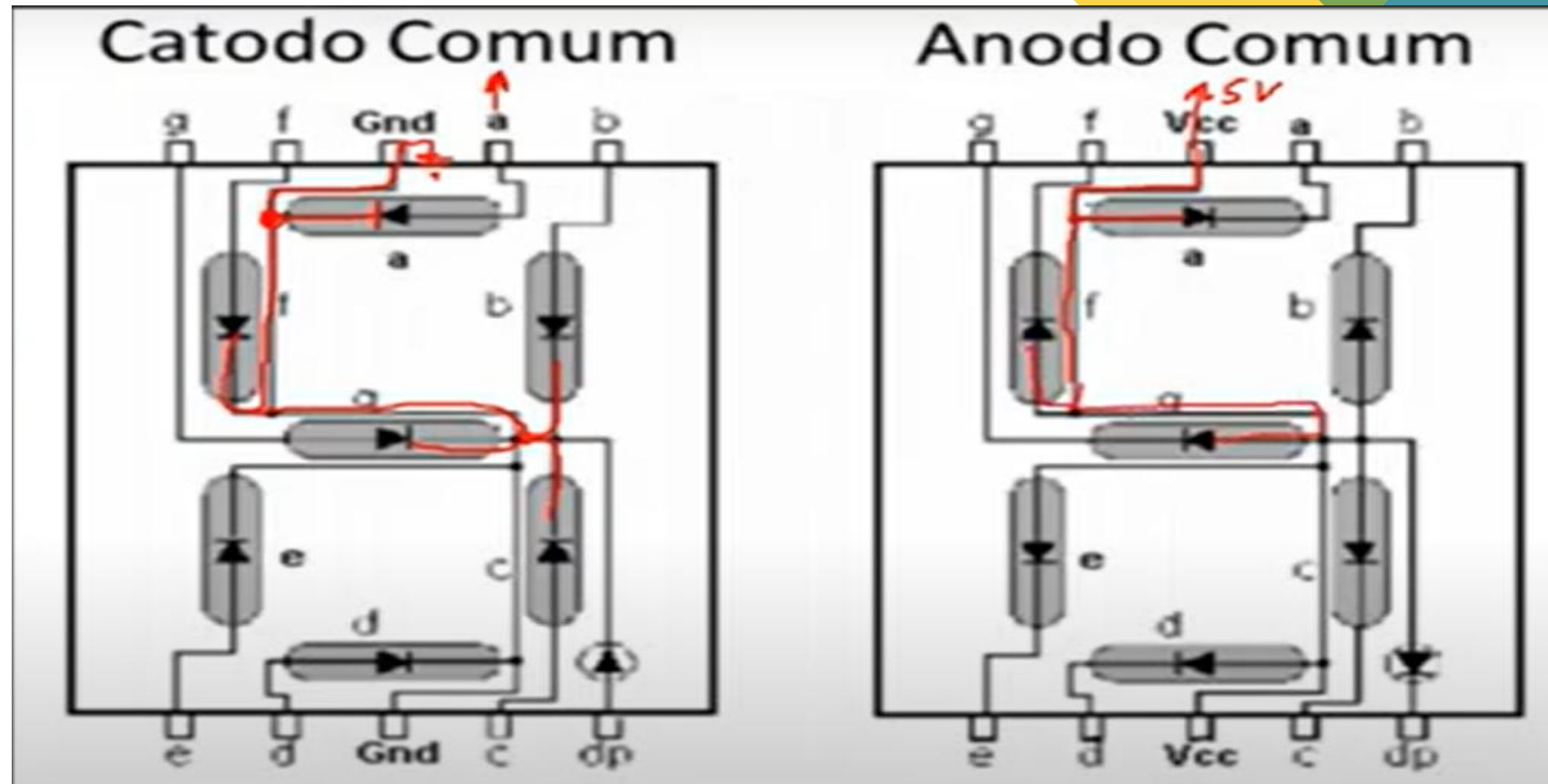


# Ánodo e Cátodo

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# Configuração



# Resistência – limitador de corrente

## ■ Absolute Maximum Rating (Ta=25°C)

| Item                       | Symbol    | Value      | Unit |
|----------------------------|-----------|------------|------|
| DC Forward Current         | $I_F$     | 100        | mA   |
| Pulse Forward Current*     | $I_{FP}$  | 1000       | mA   |
| Reverse Voltage            | $V_R$     | 5          | V    |
| Power Dissipation          | $P_D$     | 180        | mW   |
| Operating Temperature      | $T_{opr}$ | -30 ~ +85  | °C   |
| Storage Temperature        | $T_{stg}$ | -40 ~ +100 | °C   |
| Lead Soldering Temperature | $T_{sol}$ | 260°C/5sec | -    |

\* Pulse Width 100us, Duty 1/100

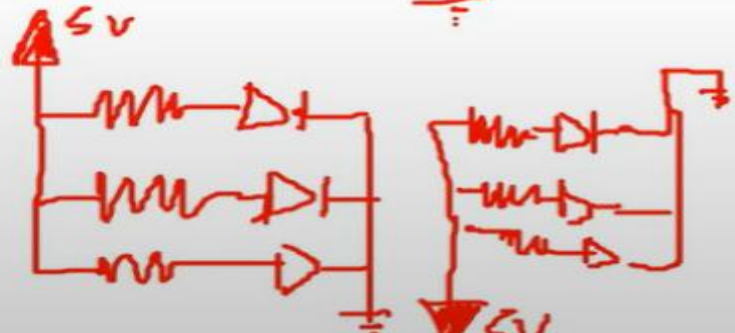
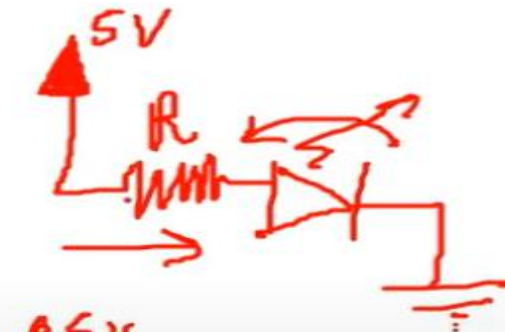
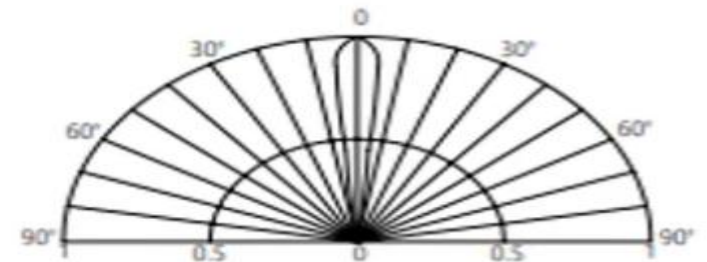
## ■ Electrical -Optical Characteristics (Ta=25°C)

| Item               | Symbol          | Condition          | Min. | Typ. | Max. | Unit          |
|--------------------|-----------------|--------------------|------|------|------|---------------|
| DC Forward Voltage | $V_F$           | $I_F=100\text{mA}$ | -    | 1.6  | 1.8  | V             |
| DC Reverse Current | $I_R$           | $V_R=5\text{V}$    | -    | -    | 10   | $\mu\text{A}$ |
| Peak Wavelength    | $\lambda_p$     | $I_F=100\text{mA}$ | -    | 940  | -    | nm            |
| Radiant Intensity* | $I_e$           | $I_F=100\text{mA}$ | -    | 35   | 45   | mW/Sr         |
| 50% Power Angle    | $2\theta_{1/2}$ | $I_F=100\text{mA}$ | -    | 15   | -    | deg           |

\*1 Tolerance of Peak wavelength is  $\pm 1\text{nm}$

\*2 Tolerance of 1 Radiant Intensity is  $\pm 15\%$

## ■ Directivity



# Cálculo da Resistência

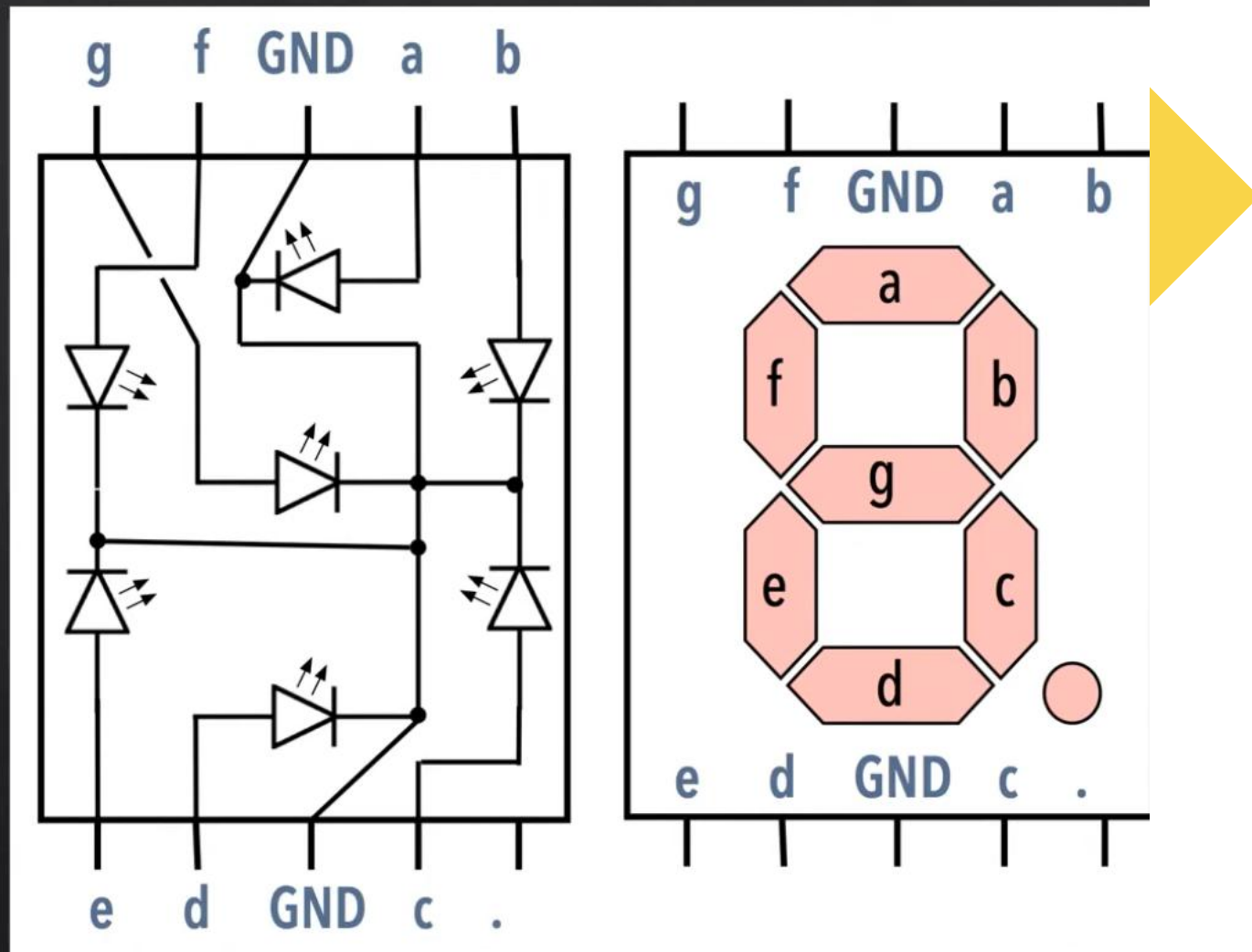
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$$R_{led} = \frac{V_{cc} - V_{led}}{I_{led}}$$

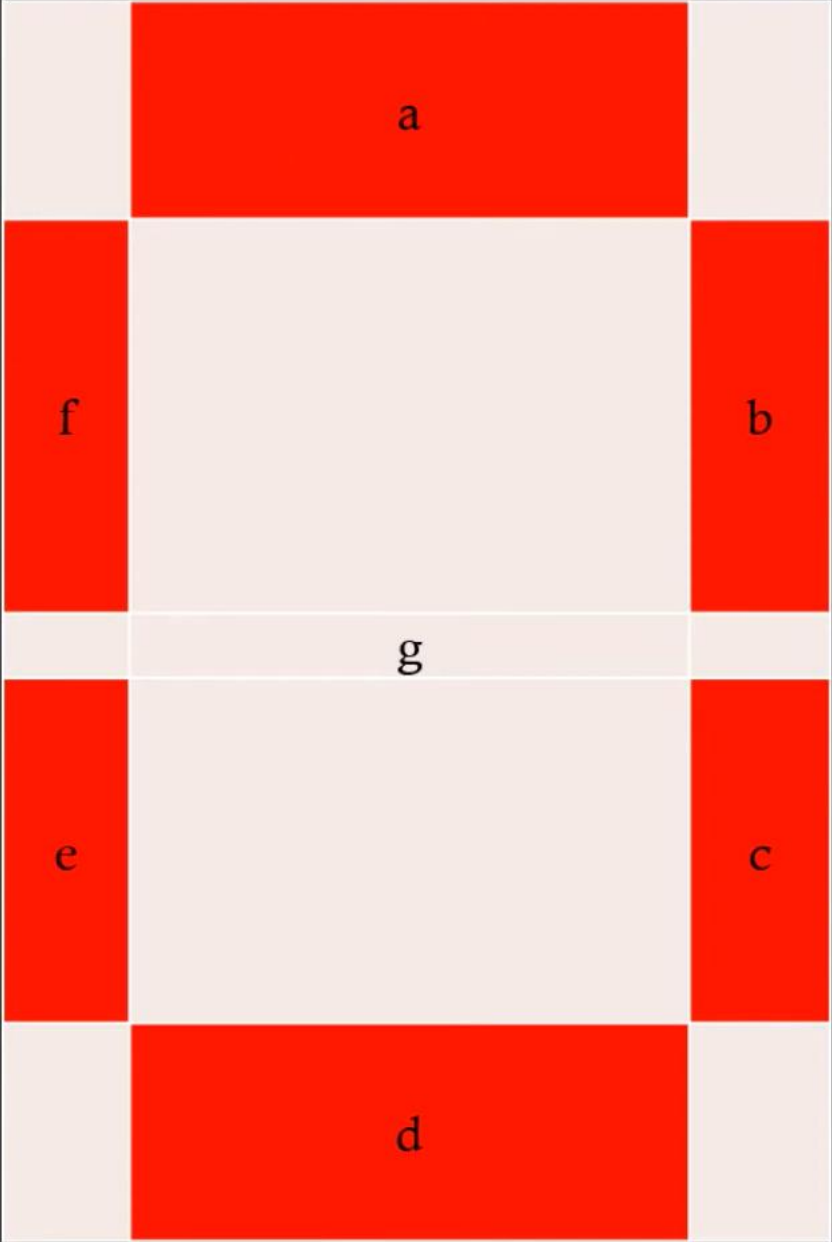
$$R_{led} = \frac{5 - 1.8}{100mA}$$

$$R_{led} = 32\Omega$$

| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  |        |   |   |   |   |   |   |
| 0        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 1  |        |   |   |   |   |   |   |



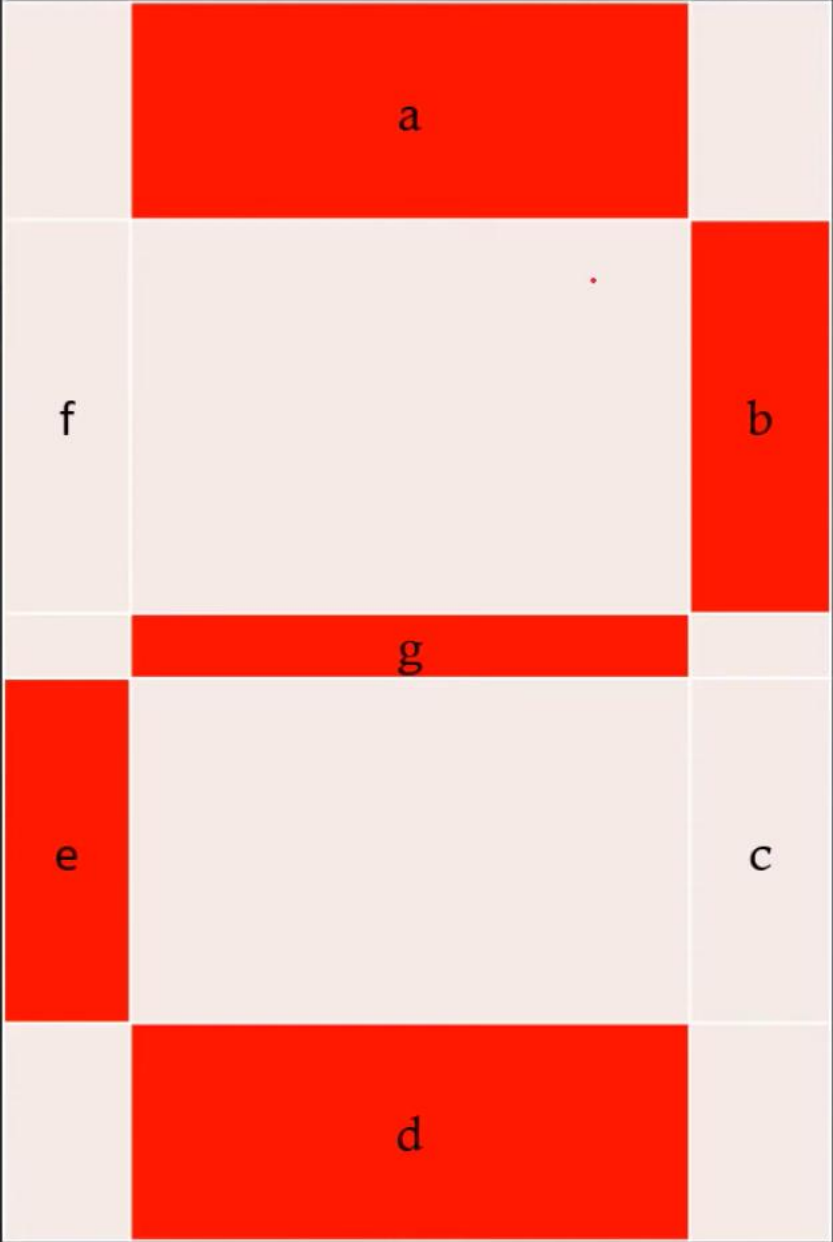
| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  |        |   |   |   |   |   |   |
| 0        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 1  |        |   |   |   |   |   |   |



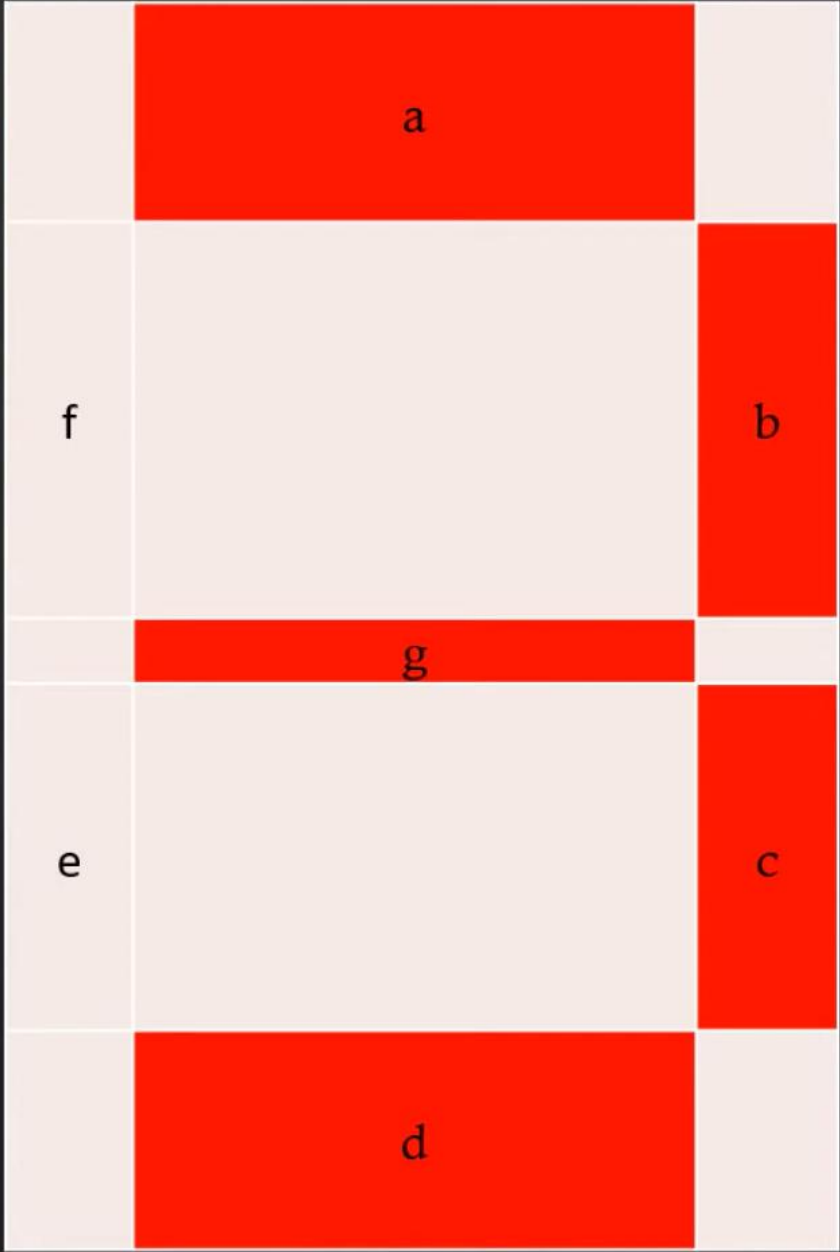
| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 1  |        |   |   |   |   |   |   |

|   |   |  |   |
|---|---|--|---|
|   | a |  |   |
|   |   |  |   |
| f |   |  | b |
|   | g |  |   |
| e |   |  | c |
|   | d |  |   |

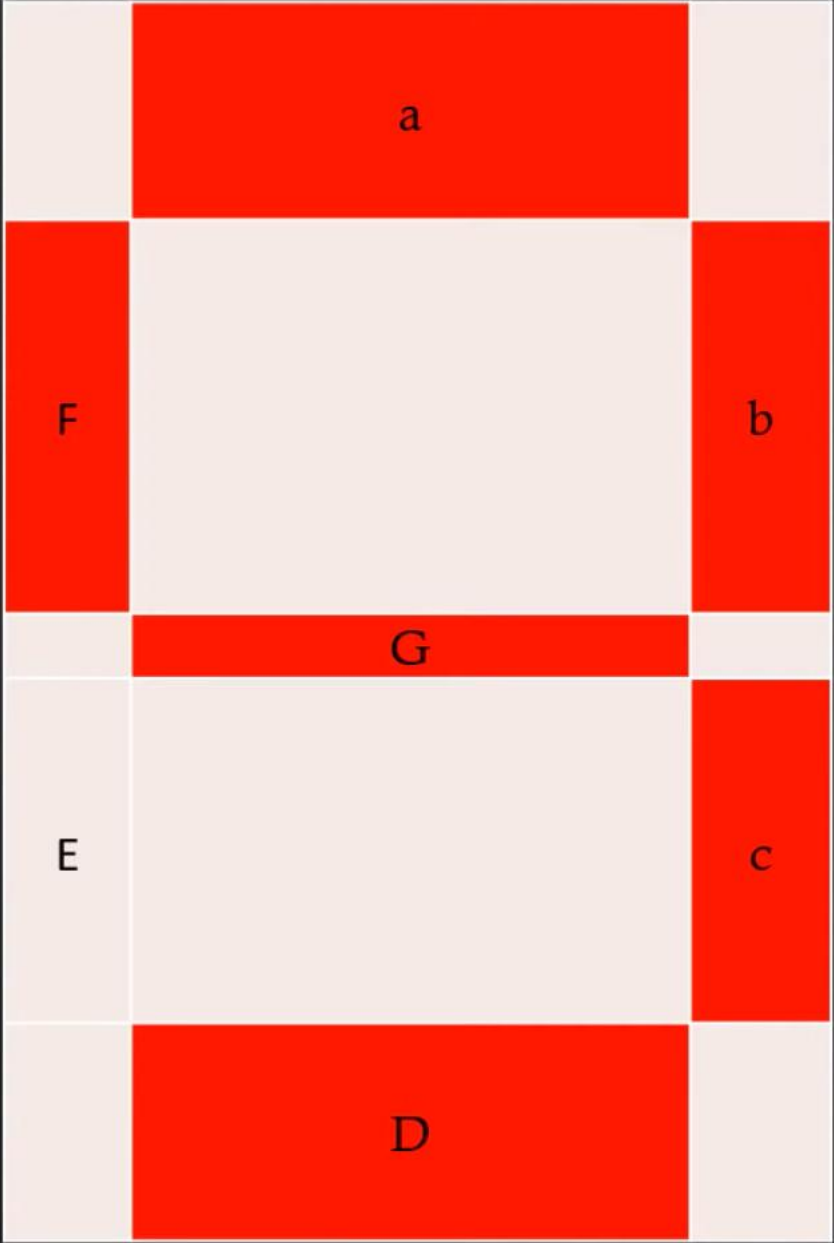
| Entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 1  |        |   |   |   |   |   |   |
|          |    |    |    |        |   |   |   |   |   |   |



| Entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 0        | 1  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 0  | 1  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 0  | 1  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 0  |        |   |   |   |   |   |   |
| 1        | 1  | 1  | 1  |        |   |   |   |   |   |   |



| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0        | 1  | 0  | 1  | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0        | 1  | 1  | 0  | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0        | 1  | 1  | 1  | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1        | 0  | 0  | 1  | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1        | 0  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 0  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |



| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0        | 1  | 0  | 1  | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0        | 1  | 1  | 0  | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0        | 1  | 1  | 1  | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1        | 0  | 0  | 1  | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1        | 0  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 0  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |

| B1 B2 |    |    |    |    |    |
|-------|----|----|----|----|----|
|       |    | 00 | 01 | 11 | 10 |
| B3 B4 | 00 | 1  | 0  | X  | 1  |
|       | 01 | 0  | 1  | X  | 1  |
|       | 11 | 1  | 1  | X  | X  |
|       | 10 | 1  | 1  | X  | X  |

| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0        | 1  | 0  | 1  | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0        | 1  | 1  | 0  | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0        | 1  | 1  | 1  | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1        | 0  | 0  | 1  | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1        | 0  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 0  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |

| B1 B2 |    |    |    |    |    |
|-------|----|----|----|----|----|
|       |    | 00 | 01 | 11 | 10 |
| B3 B4 | 00 | 1  | 0  | X  | 1  |
|       | 01 | 0  | 1  | X  | 1  |
|       | 11 | 1  | 1  | X  | X  |
|       | 10 | 1  | 1  | X  | X  |

$$a = B_1 + B_3 + (B_2 \odot B_4)$$

| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0        | 1  | 0  | 1  | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0        | 1  | 1  | 0  | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0        | 1  | 1  | 1  | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1        | 0  | 0  | 1  | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1        | 0  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 0  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |

| B1 B2 |    |    |    |    |    |
|-------|----|----|----|----|----|
|       |    | 00 | 01 | 11 | 10 |
| B3 B4 | 00 | 1  | 1  | X  | 1  |
|       | 01 | 1  | 0  | X  | 1  |
|       | 11 | 1  | 1  | X  | X  |
|       | 10 | 1  | 0  | X  | X  |

| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0        | 1  | 0  | 1  | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0        | 1  | 1  | 0  | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0        | 1  | 1  | 1  | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1        | 0  | 0  | 1  | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1        | 0  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 0  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |

| B1 B2 |    |    |    |    |    |
|-------|----|----|----|----|----|
|       |    | 00 | 01 | 11 | 10 |
| B3 B4 | 00 | 1  | 1  | X  | 1  |
|       | 01 | 1  | 0  | X  | 1  |
|       | 11 | 1  | 1  | X  | X  |
|       | 10 | 1  | 0  | X  | X  |

$$b = \overline{B_2} + (B_3 \odot B_4)$$

| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0        | 1  | 0  | 1  | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0        | 1  | 1  | 0  | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0        | 1  | 1  | 1  | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1        | 0  | 0  | 1  | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1        | 0  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 0  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |

| B1 B2 |    |    |    |    |    |
|-------|----|----|----|----|----|
|       |    | 00 | 01 | 11 | 10 |
| B3 B4 | 00 | 1  | 1  | X  | 1  |
|       | 01 | 1  | 1  | X  | 1  |
|       | 11 | 1  | 1  | X  | X  |
|       | 10 | 0  | 1  | X  | X  |

C =

| entradas |    |    |    | saídas |   |   |   |   |   |   |
|----------|----|----|----|--------|---|---|---|---|---|---|
| B1       | B2 | B3 | B4 | a      | b | c | d | e | f | g |
| 0        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0        | 0  | 0  | 1  | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0        | 0  | 1  | 0  | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0        | 0  | 1  | 1  | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0        | 1  | 0  | 0  | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0        | 1  | 0  | 1  | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0        | 1  | 1  | 0  | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0        | 1  | 1  | 1  | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1        | 0  | 0  | 0  | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1        | 0  | 0  | 1  | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1        | 0  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 0  | 1  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 0  | 1  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 0  | x      | x | x | x | x | x | x |
| 1        | 1  | 1  | 1  | x      | x | x | x | x | x | x |

| B1 B2 |    |    |    |    |    |
|-------|----|----|----|----|----|
| B3 B4 |    | 00 | 01 | 11 | 10 |
|       | 00 | 1  | 1  | X  | 1  |
|       | 01 | 1  | 1  | X  | 1  |
|       | 11 | 1  | 1  | X  | X  |
|       | 10 | 0  | 1  | X  | X  |

$$C = \overline{B_3} + B_4 + B_2$$

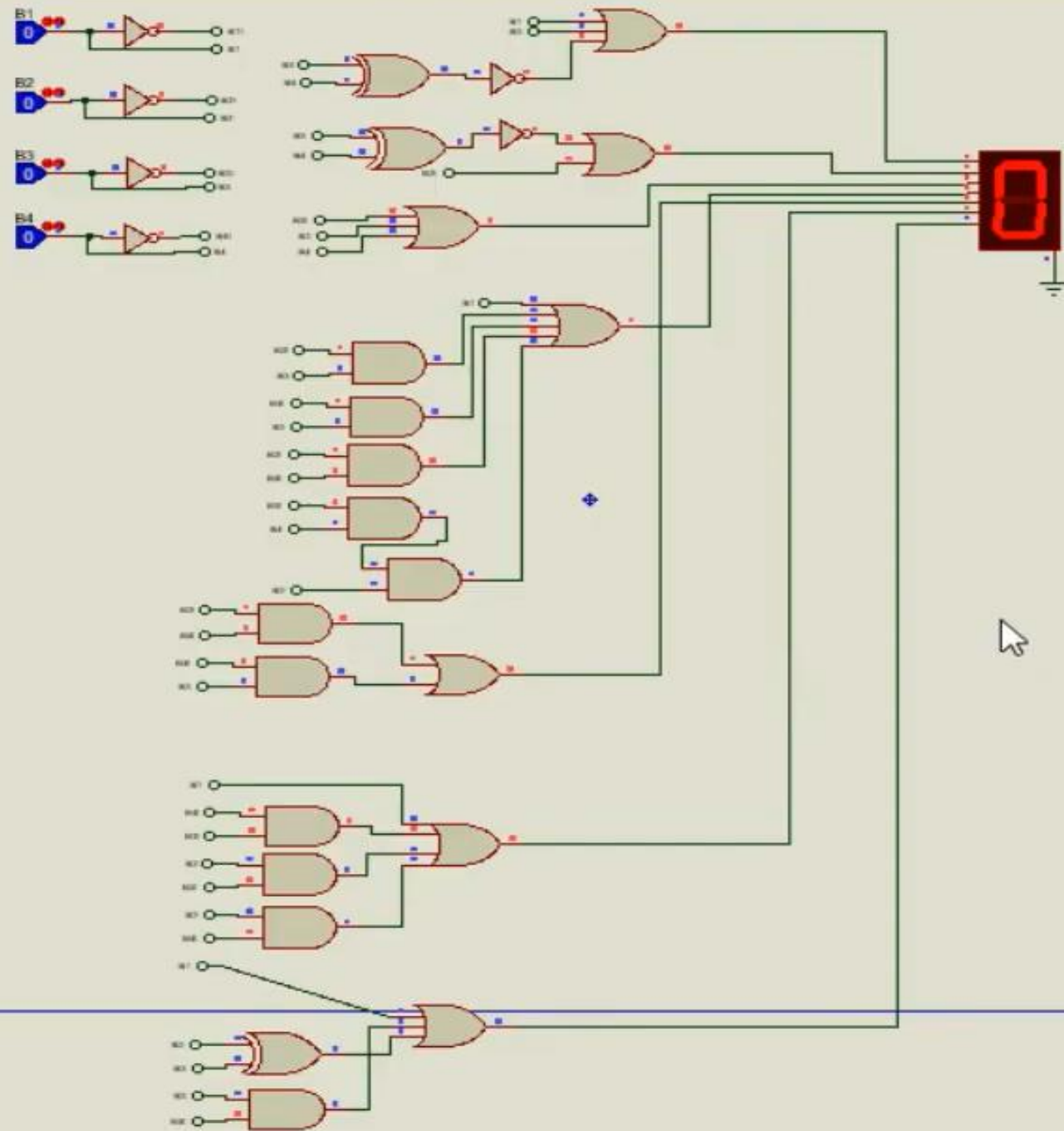
## Outros sinais

$$d = B_1 + B_3 \cdot \overline{B_4} + \overline{B_2} \cdot B_3 + \overline{B_2} \cdot \overline{B_4} + B_2 \cdot \overline{B_3} \cdot B_4$$

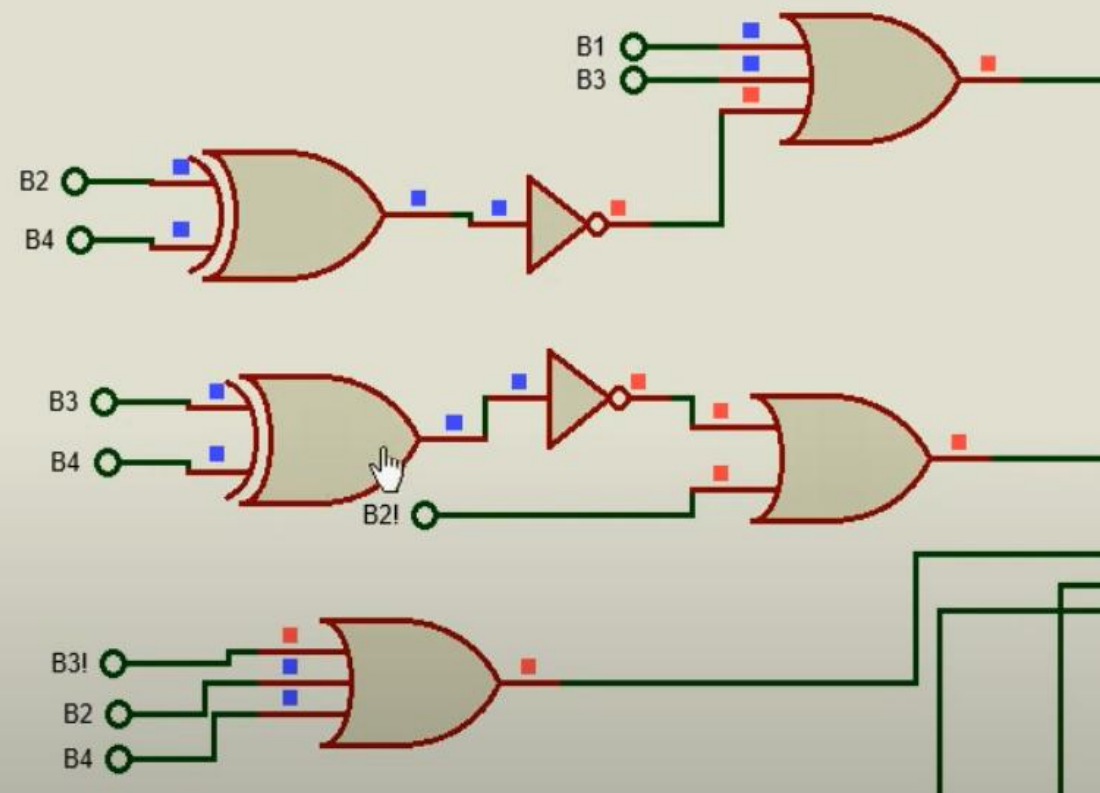
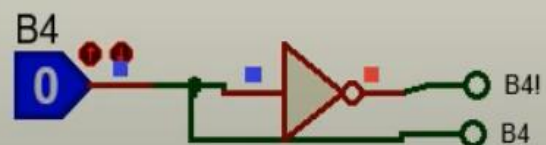
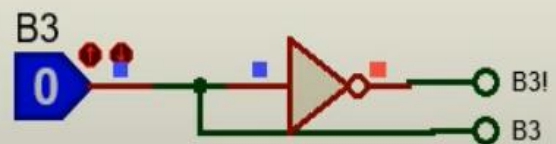
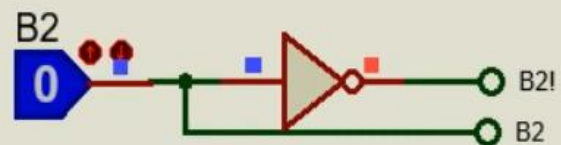
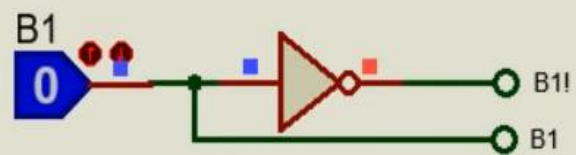
$$e = \overline{B_2} \cdot \overline{B_4} + B_3 \cdot \overline{B_4}$$

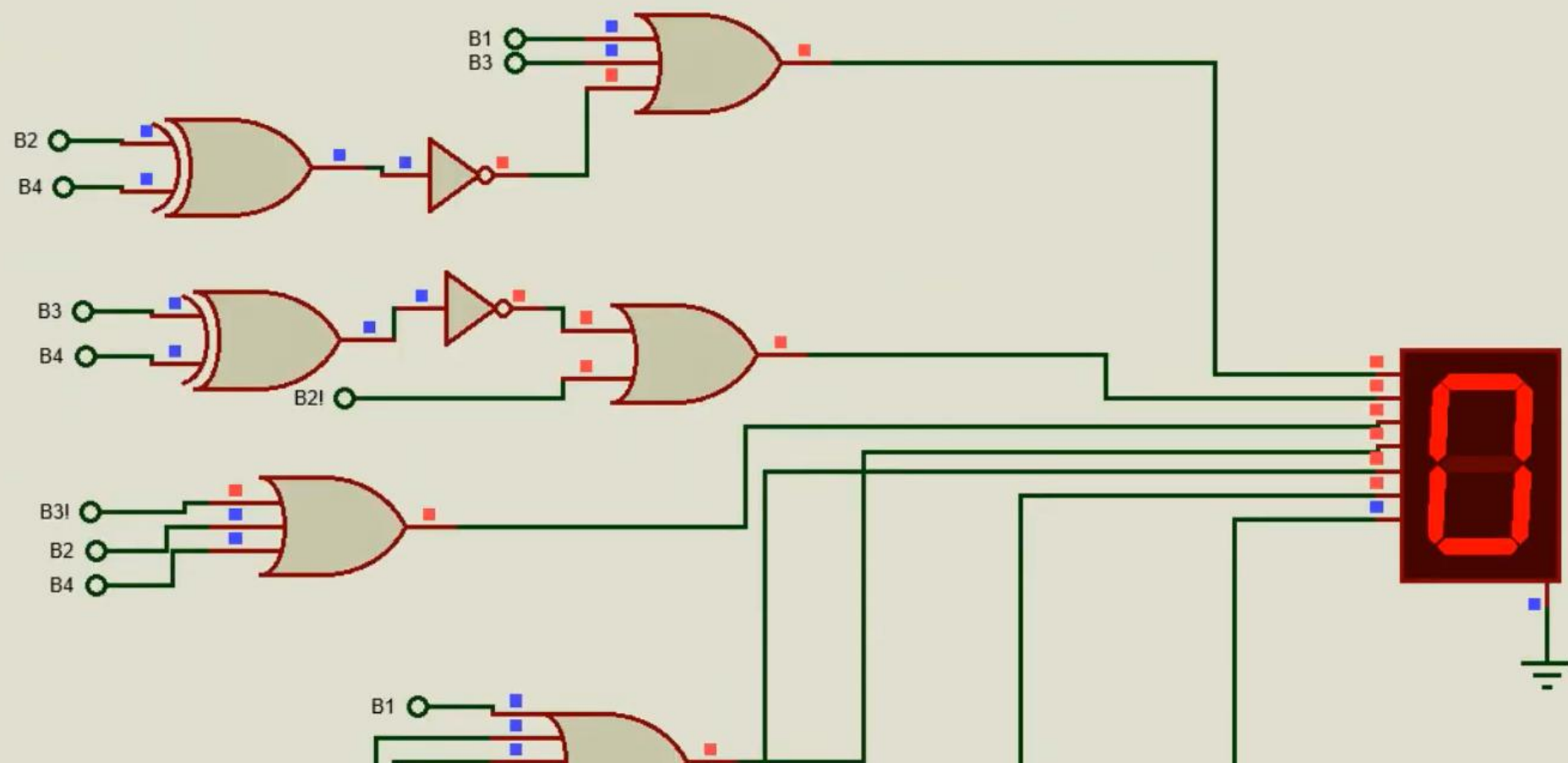
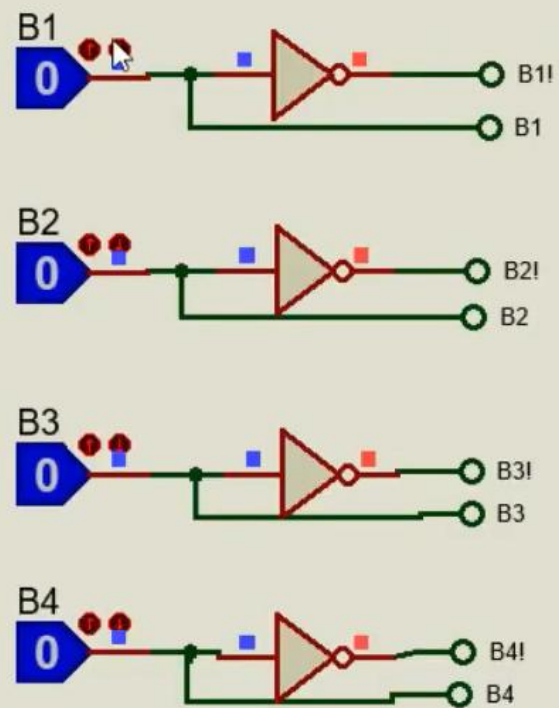
$$f = \overline{B_3} \cdot \overline{B_4} + B_2 \cdot \overline{B_3} + B_2 \cdot \overline{B_4}$$

$$g = B_1 + B_2 \cdot \overline{B_4} + (B_2 \oplus B_3)$$

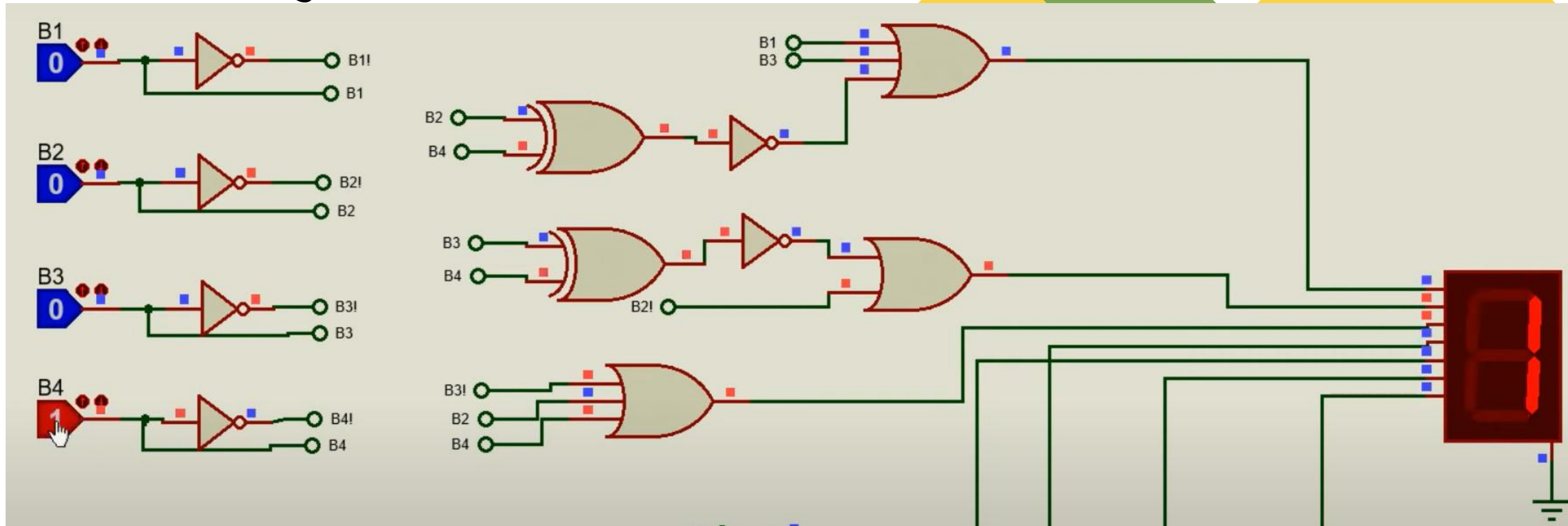


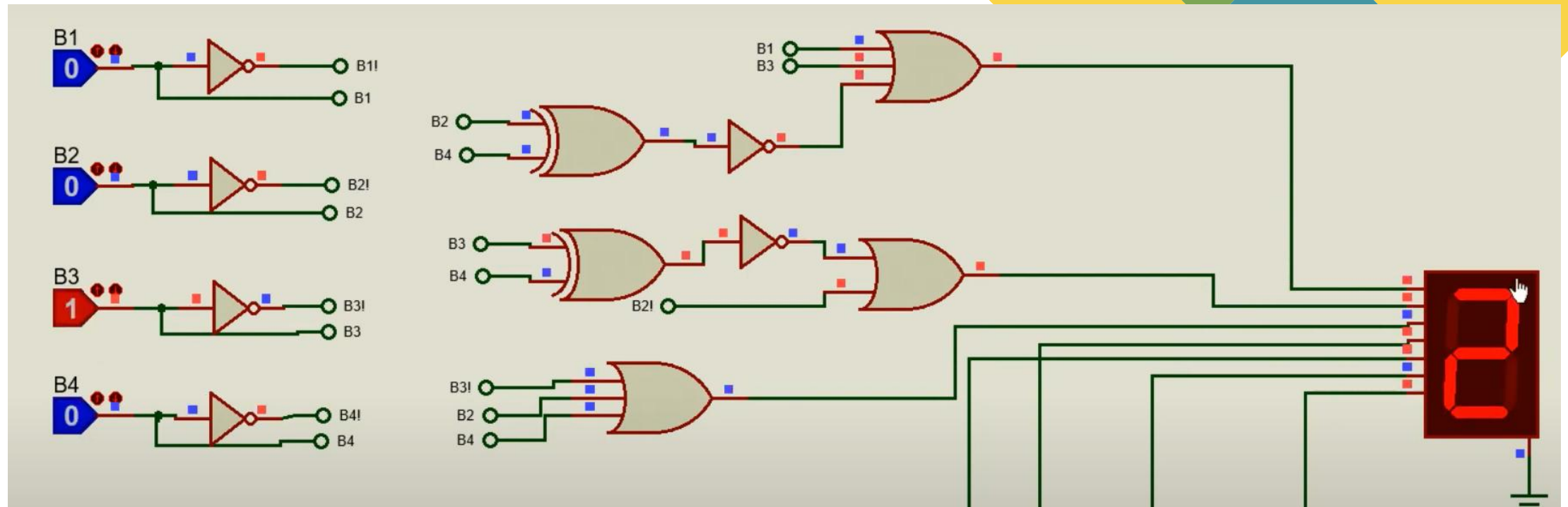
# tags

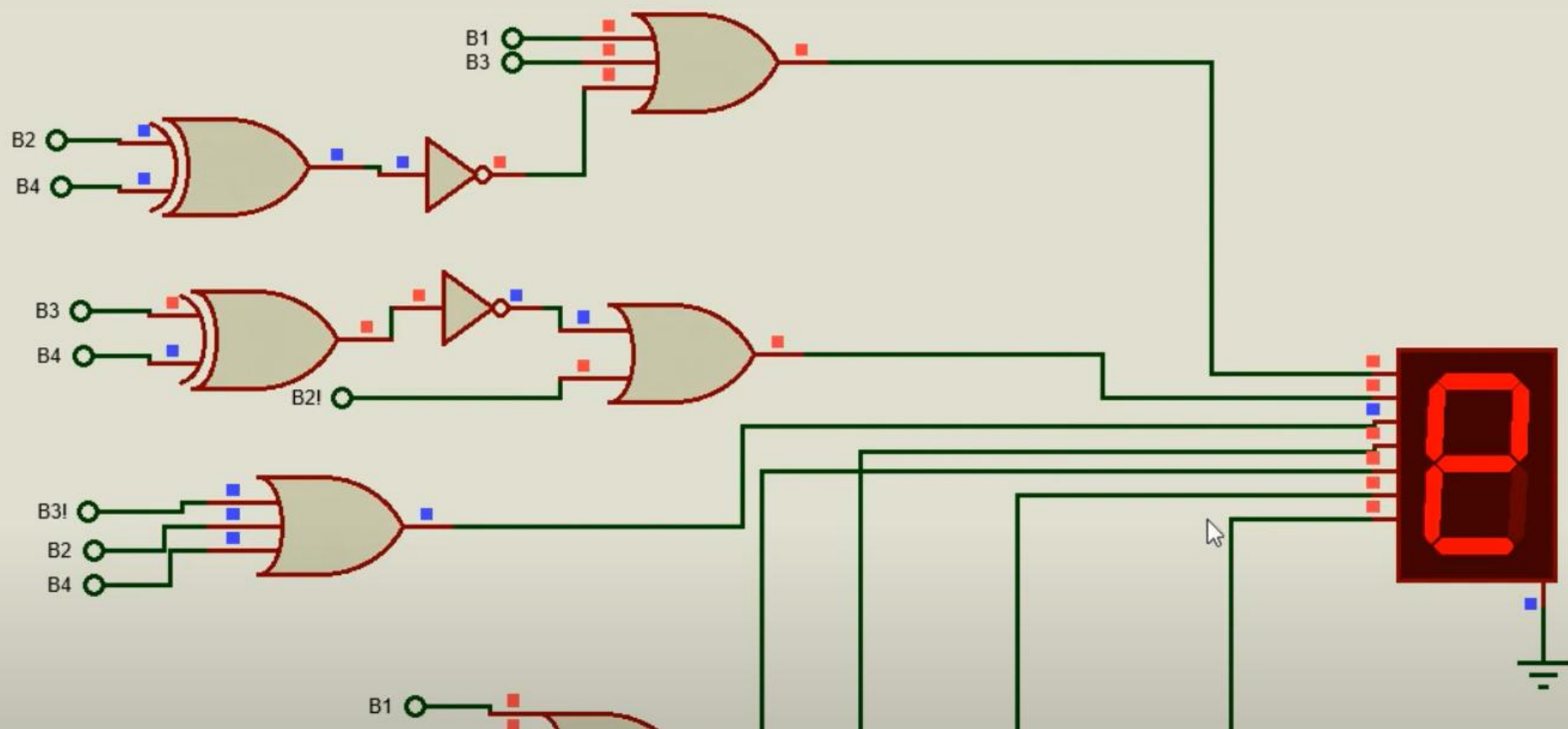
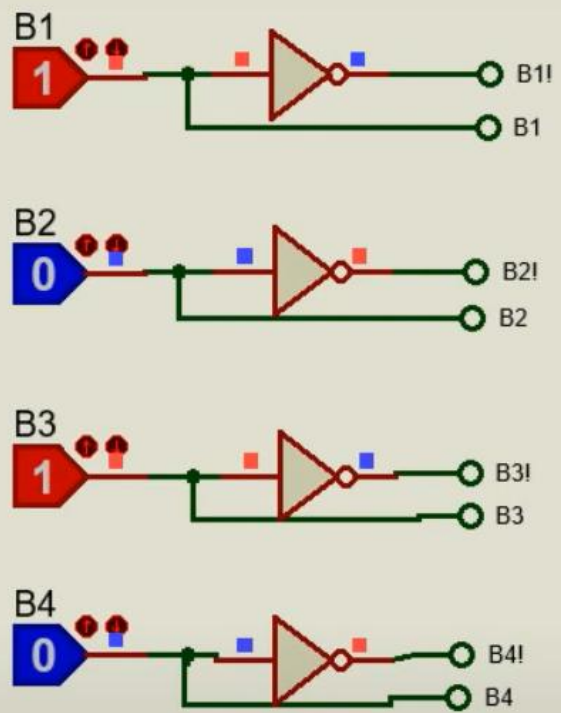




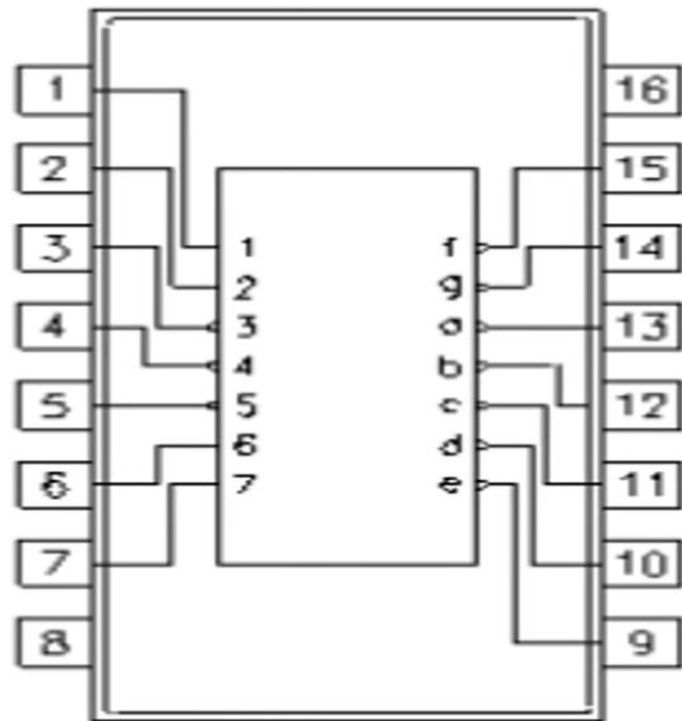
# Simulação





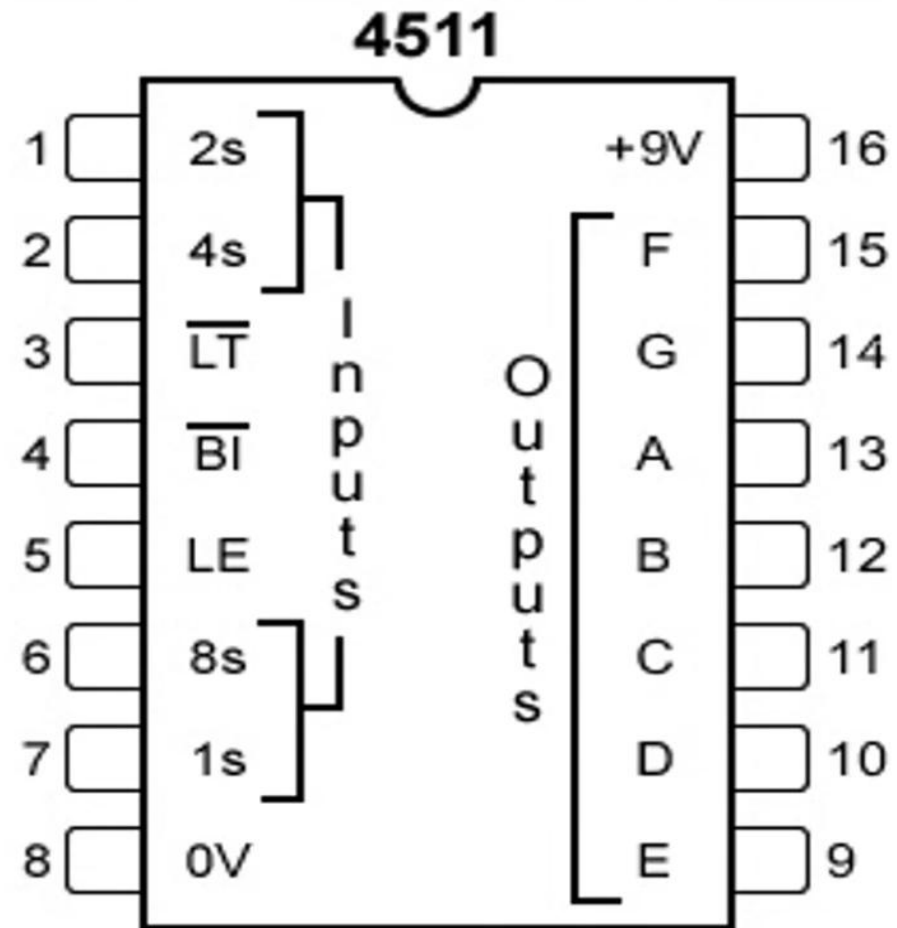


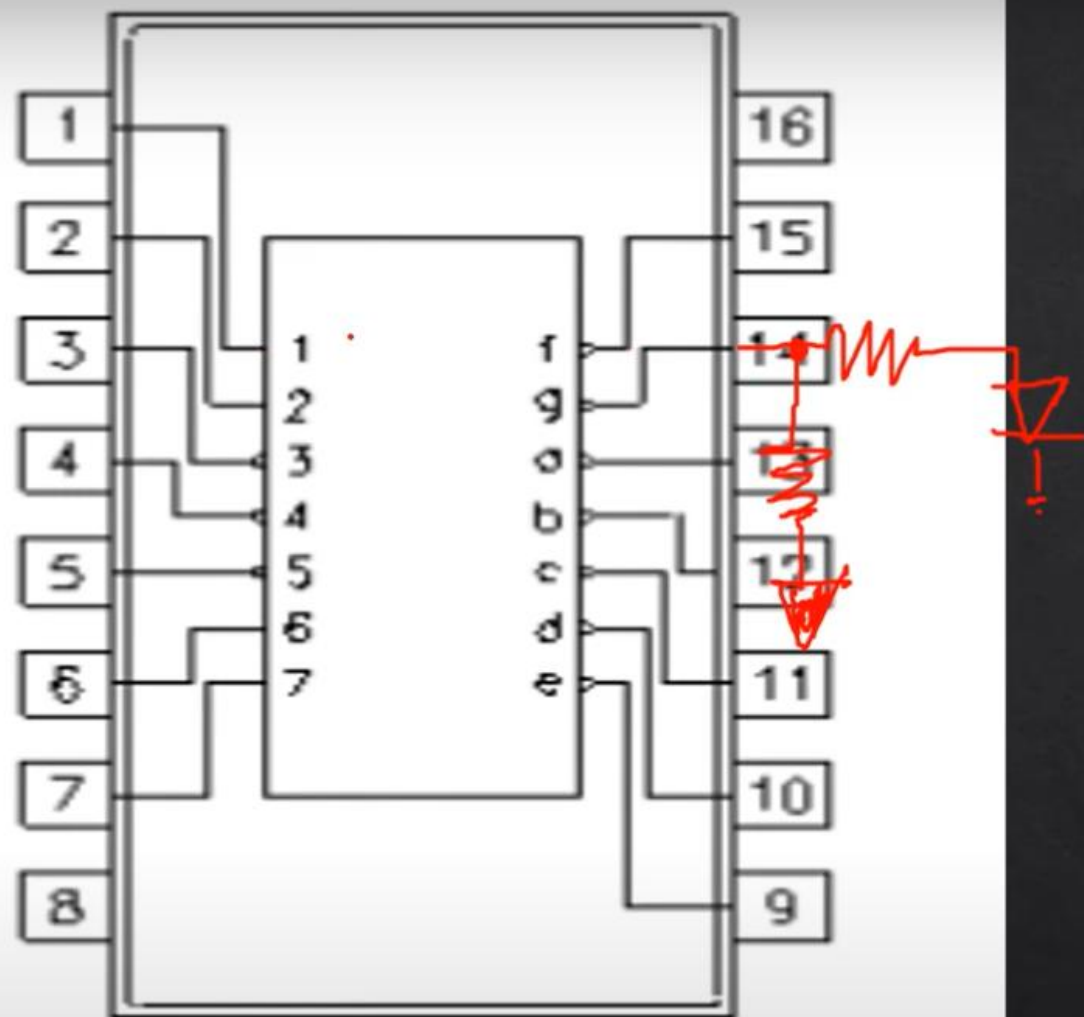
# Decodificadores BCD para 7 segmentos



7446

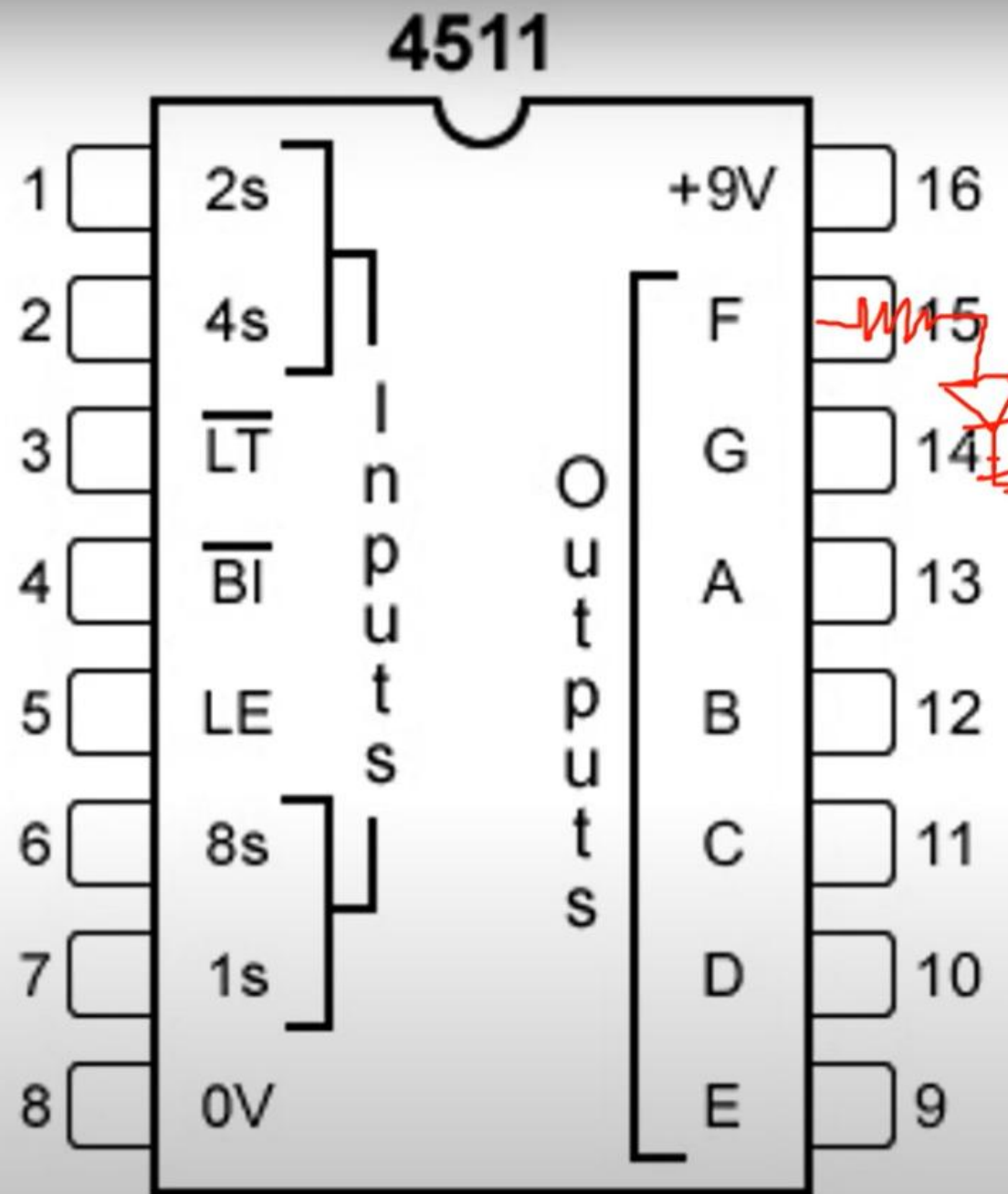
BCD to 7-Segment  
Decoder/Driver with (30V)  
Open Collector Output





7446

BCD to 7-Segment  
Decoder/Driver with (30V)  
Open Collector Output



# Visão Geral

---

- **7446**

- Decodificador/driver BCD para display de 7 segmentos com saídas open-collector, ativo em nível baixo, para displays de ânodo comum

- **CD4511**

- Decodificador/latch/driver BCD para display de 7 segmentos com saídas ativas em nível alto, para displays de cátodo comum.

# Arquitetura interna (conceitual)



---

- **7446**
  - Não possui latch interno
  - Focado em lógica TTL discreta
- **4511**
  - Possui memória (latch)
  - Mais “inteligente” e moderno

# Funcionamento elétrico

---

- **7446**
  - Saída funciona como um interruptor para o GND
  - Não fornece tensão positiva
- **Para acender:**
  - corrente flui:  $V_{cc} \rightarrow \text{display} \rightarrow \text{CI} \rightarrow \text{GND}$
- **Excelente para:**
  - cargas maiores
  - integração com outros níveis



# Funcionamento elétrico

---

- Saída fornece nível HIGH diretamente
- Para acender:
  - saída fornece corrente ao LED
- Mais simples de usar
- Menos componentes externos



# Entradas

---

- 7446
  - LT (Lamp Test) → acende tudo
  - BI (Blanking Input) → apaga
  - RBI / RBO → supressão de zeros
  - Muito usado em displays múltiplos



# Aplicações

---

- 7446
  - Sistemas TTL clássicos
  - Contadores discretos (ex: 7490)
  - Painéis industriais
  - Displays grandes



# Aplicações

---

- 4511
  - Projetos didáticos
  - Sistemas CMOS
  - Interfaces com microcontroladores
  - Protótipos rápidos



# Vantagens e desvantagens



---

- 7446
  - Suporta maior corrente (sink)
  - Ideal para lógica TTL
  - Permite cascata (RBI/RBO)
  - Circuito mais complexo
  - Não tem latch



---

- **4511**

- Fácil de usar
- Tem latch interno
- Menor capacidade de corrente
- Não ideal para cargas maiores

# Comparação prática

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- Comparação prática (intuição)
- Pense assim:
  - 7446 = “chave para o terra”
  - 4511 = “fonte que acende o LED”





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- **4511**

- LE (Latch Enable) → memória
- LT (Lamp Test)
- BI (Blanking Input)

- **Melhor para sistemas controlados/microprocessados**

# Atividade

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- Projetar um codificador BCD para um display de 7 segmentos (números 0-9, A, B, C, D, E.
- Display Anodo

